ViGoEmotions: Task and Dataset

Multi-label emotion recognition in Vietnamese comments.

20,664 comments; 27 emotions plus a Neutral state.

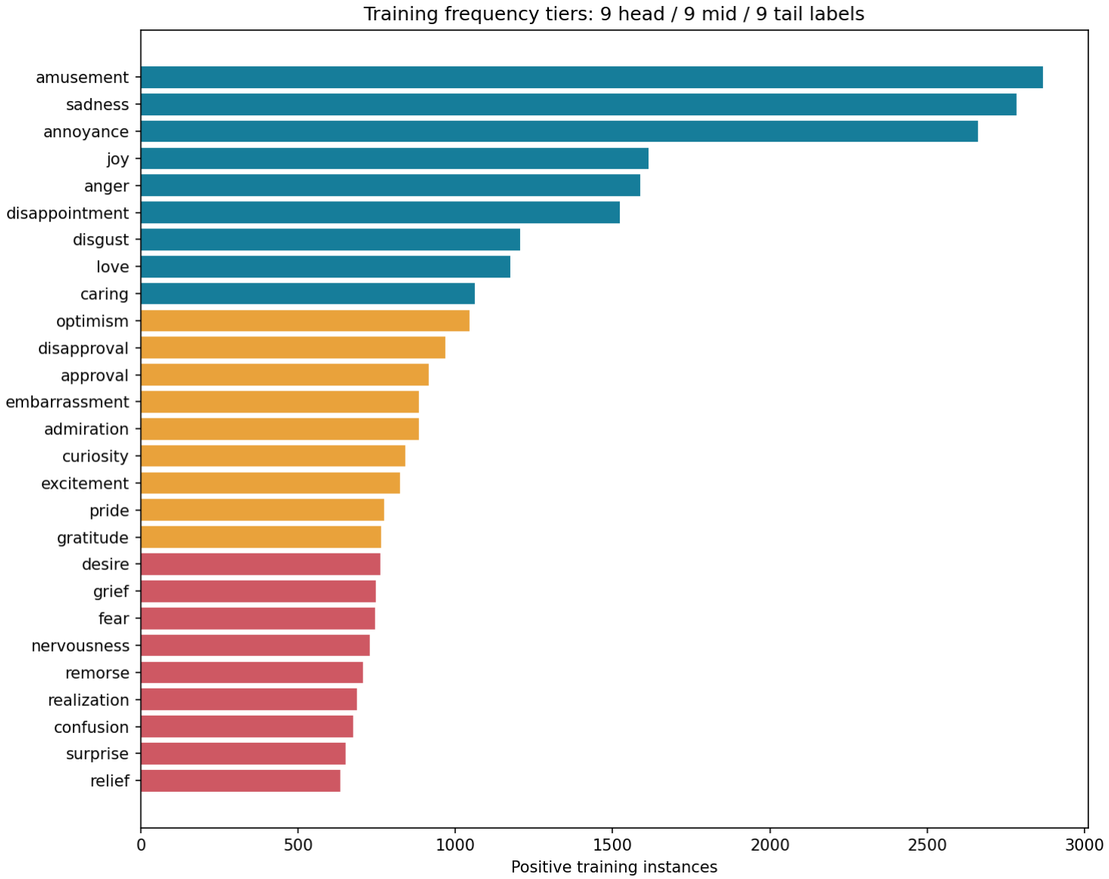
Effective train / validation / test: 16,530 / 2,066 / 2,067.

Three models; compare 27 and 28 outputs.

All model scores are validation results.

Data Profile and Leakage Prevention

- Mean 1.859 emotions; median 11 tokens / 47 characters.
- Amusement–relief frequency ratio: about 4.52×.
- Fixed split; one overlapping training sample removed.
- Fit vocabulary/IDF on train only; no model test scores.



27 vs 28 Outputs: What Changes?

27 outputs: Neutral-only has an all-zero emotion target.

28 outputs: Neutral becomes a separate learned label.

Keep mixed annotations; no Neutral exclusion rule.

Validation: 75 Neutral-only + 19 mixed = 94.

Independent sigmoid outputs with binary cross-entropy.

Models and Shared Training Protocol

TF-IDF + OvR LR: liblinear; max_iter = 1,000.

BiGRU from scratch: hidden 128; batch 64; lr 0.001.

PhoBERT full fine-tuning: batch 16; lr 2e-5.

Both neural models: 20 epochs per label setting; seed 42.

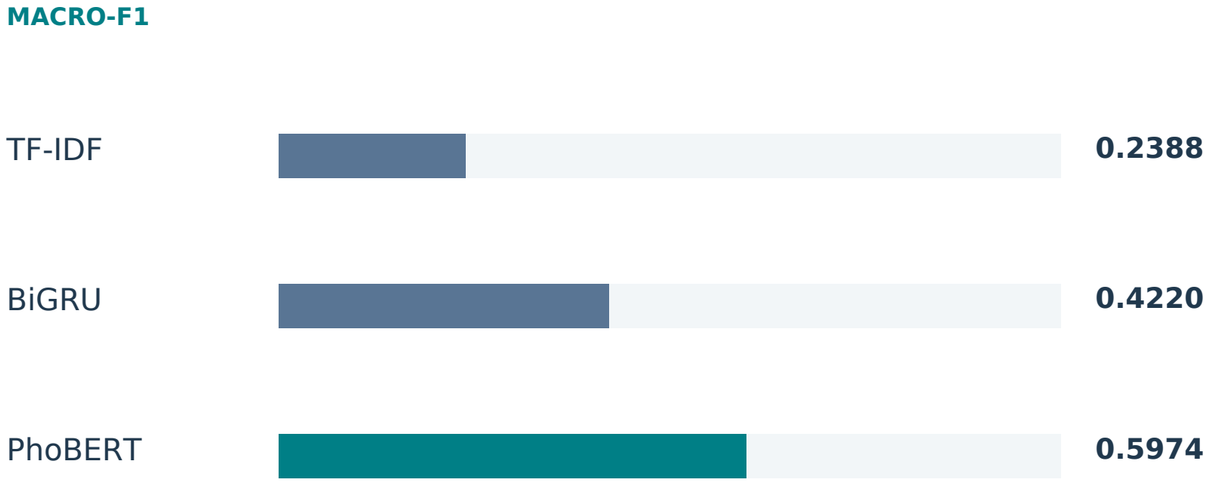
Select checkpoints by Macro-F1 on the same 27 emotions.

Results: 27 Emotions

TF-IDF Macro-F1: 0.2388.

BiGRU Macro-F1: 0.4220.

PhoBERT Macro-F1: 0.5974; Micro-F1: 0.5996.



PhoBERT Hamming Loss: 0.0550; LRAP: 0.7419.

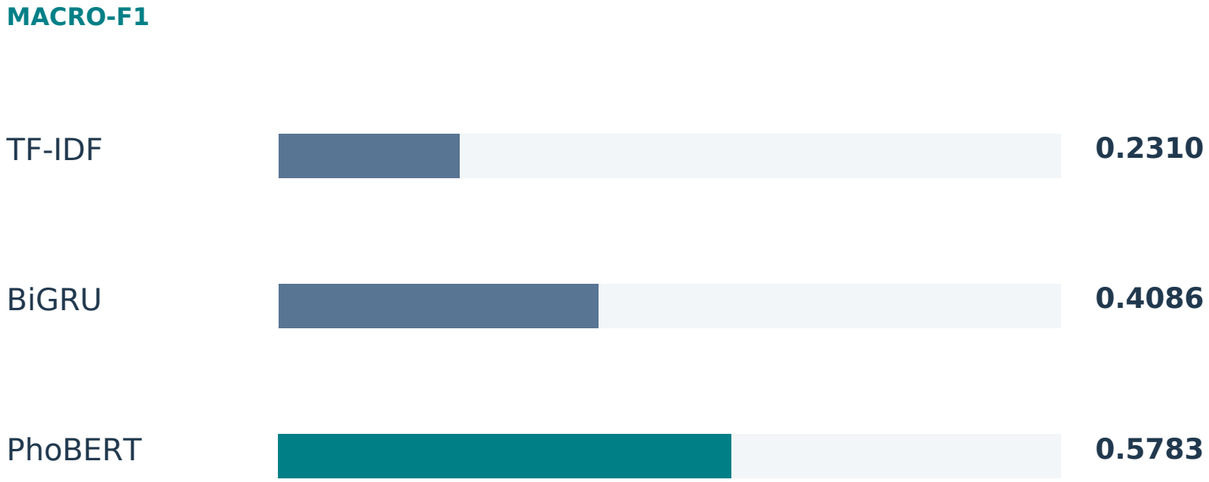
Higher F1/LRAP is better; lower Hamming Loss is better.

Results: 28 Labels Including Neutral

TF-IDF Macro-F1: 0.2310.

BiGRU Macro-F1: 0.4086.

PhoBERT Macro-F1: 0.5783; Micro-F1: 0.5829.



PhoBERT Hamming Loss: 0.0575; LRAP: 0.7077.

These metrics evaluate all 28 labels.

Does Neutral Help the Shared 27 Emotions?

PhoBERT with 27 outputs: 0.5974.

PhoBERT with 28 outputs, scored on 27: 0.5836.

Difference: -1.38 percentage points.

No observed improvement on the shared emotions.

One seed; no causal or statistical conclusion.

Neutral Detection Remains Difficult

PhoBERT Neutral F1: 0.4369.

Precision: 0.4018; recall: 0.4787.

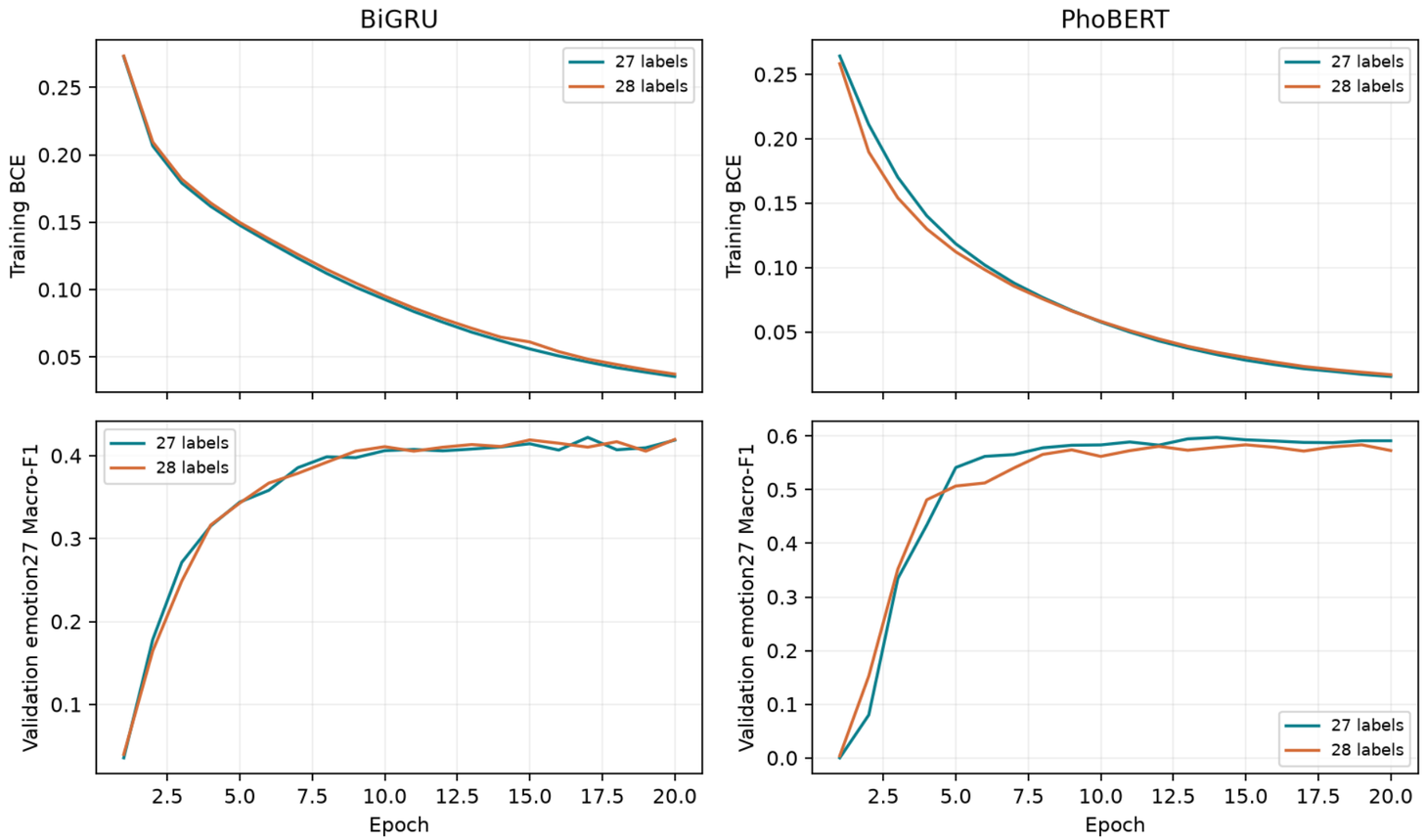
45 TP / 67 FP / 49 FN; support: 94.

The 27-output model has no Neutral classifier.

An empty prediction is not automatically Neutral.

Learning Curves and Best Checkpoints

- All four neural runs completed epochs 1–20.
- Best BiGRU epochs, 27/28: 17 / 20.
- Best PhoBERT epochs, 27/28: 14 / 15.
- Select the best checkpoint, not necessarily the last.
- Training loss alone cannot confirm overfitting.



Q1–Q2: Detailed Result Tables

27-output study only · validation · threshold 0.5 · dedicated tables for Q1 and Q2

Q1 / HIGHEST AND LOWEST PER-LABEL F1

Model	Highest label / F1	Lowest label / F1
TF-IDF + OvR LR	gratitude 0.6797	realization 0.0000
BiGRU	gratitude 0.7701	realization 0.0940
PhoBERT	**gratitude 0.8370**	**disapproval 0.3623**

Q2 / FREQUENCY–F1 CORRELATION

Model	Spearman ρ	p-value
TF-IDF + OvR LR	0.242	0.223
BiGRU	0.313	0.112
PhoBERT	**0.018**	0.928

Q1 reports each model’s highest and lowest validation F1. Q2 correlates training frequency with per-label validation F1 across 27 emotions. Descriptive, one-seed analysis; no causal or significance claim.

Sources: current label_extremes.csv; frequency_f1.csv. All results use 27 outputs; no test metrics.

RESEARCH QUESTION TABLES

Q3–Q4: Pair and Text-Style Tables

27-output study only · validation · threshold 0.5 · dedicated tables for Q3 and Q4

Q3 / DISAPPOINTMENT + SADNESS · TRAIN N = 904

Model	Joint TP	Support	Recall
TF-IDF + OvR LR	2	122	1.64%
BiGRU	37	122	30.33%
PhoBERT	**63**	122	**51.64%**

Q4 / TEXT-STYLE MACRO-F1

Group	n	BiGRU F1	PhoBERT F1	Δ (pp)
All	2,066	0.4220	0.5974	+17.54
Emoji-heavy	199	0.3506	0.4789	+12.83
Short	343	0.4303	0.5654	+13.51
Teencode	307	0.3333	0.5547	**+22.13**

Joint recall counts both emotions predicted among examples annotated with both; extra predicted labels are allowed. Style groups overlap and differ in size; Δ = PhoBERT − BiGRU in Macro-F1 percentage points. One seed; no causal or significance claim.

Sources: current pair_preservation.csv; neural_style_comparison.csv. Pair selected by training co-occurrence; no test metrics.

Q5: Does the Default Threshold Matter?

PhoBERT 27 at 0.5: Macro-F1 0.5974.

Best validation grid point: 0.35, F1 0.5978.

Same checkpoint; exploratory validation analysis.

Very small F1 gain; Hamming Loss increases.

Main results remain at threshold 0.5.

Error Analysis: New Neutral Predictions

Six cases: three Neutral false positives and three misses.



Potential issues: implied emotion, short text, HTML, profanity.

IDs and abstract observations; no raw comments.

Preliminary AI-assisted review; team confirmation pending.

Final Research Question and Next Steps

Is one threshold suitable for every emotion?

Freeze the best neural checkpoint and its weights. → Compare 0.5, global tuned and per-label thresholds. → Tune on validation; lock choices; evaluate test once.

Review at least 20 false negatives and 20 false positives.

Conclusions and Evidence

PhoBERT leads within both label settings.

Neutral does not improve shared-emotion F1 in this run.

Evidence: Kaggle 08/09, saved predictions and manifests.

Limits: one seed, validation selection, no test scores.

Sources: course brief, ViGoEmotions, PhoBERT and libraries.